

TCGA-MeSH Mapping and Literature-Derived GRN Analysis

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Objective

This report summarises the current literature-derived context-dependent GRN analysis for TCGA-mapped cancer types. TCGA cancer types were mapped to one or more corresponding MeSH disease descriptors, from which disease-specific network-derived gene rankings were generated. PCA and UMAP were subsequently used to provide an overview of similarities among the mapped disease networks.

Analysis summary

A curated TCGA-to-MeSH mapping was constructed for all cancer types included in this study. Where a single MeSH descriptor adequately represented a TCGA cancer type, a one-to-one mapping was used. Where anatomical site and histological subtype were both important, multiple MeSH descriptors were retained to preserve disease specificity.

For each selected MeSH descriptor, the corresponding literature-derived disease network was obtained from the available MeSH GRN resources. While the Zenodo dataset provided pre-generated networks for only a subset of the required MeSH descriptors, the remaining disease networks were reconstructed using the established Fig. 2-0 GRN construction workflow from the project repository. This produced a consistent set of disease-specific network profiles for downstream analysis.

Two analyses were performed from these disease networks. Firstly, genes within each disease network were ranked according to their network importance, producing a ranked gene list for every TCGA-MeSH mapping. Secondly, the disease-level network profiles were compared using PCA and UMAP, providing an exploratory overview of similarities among the mapped diseases.

The analyses presented in this report are based exclusively on the literature-derived disease networks. TCGA information is used only to define the disease mappings, and differential expression analysis will be incorporated separately at a later stage as a complementary comparison.

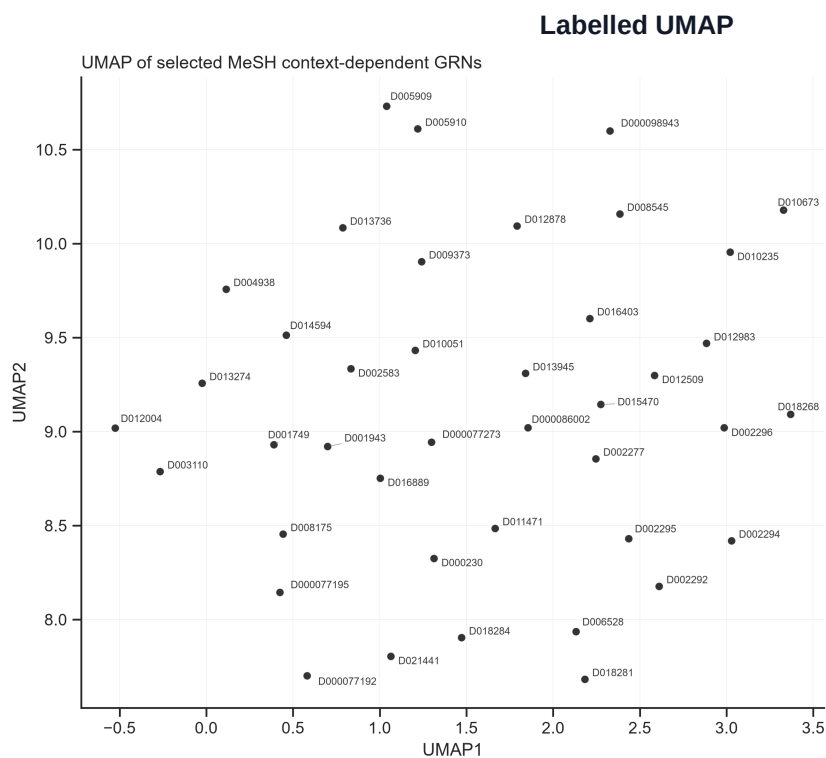
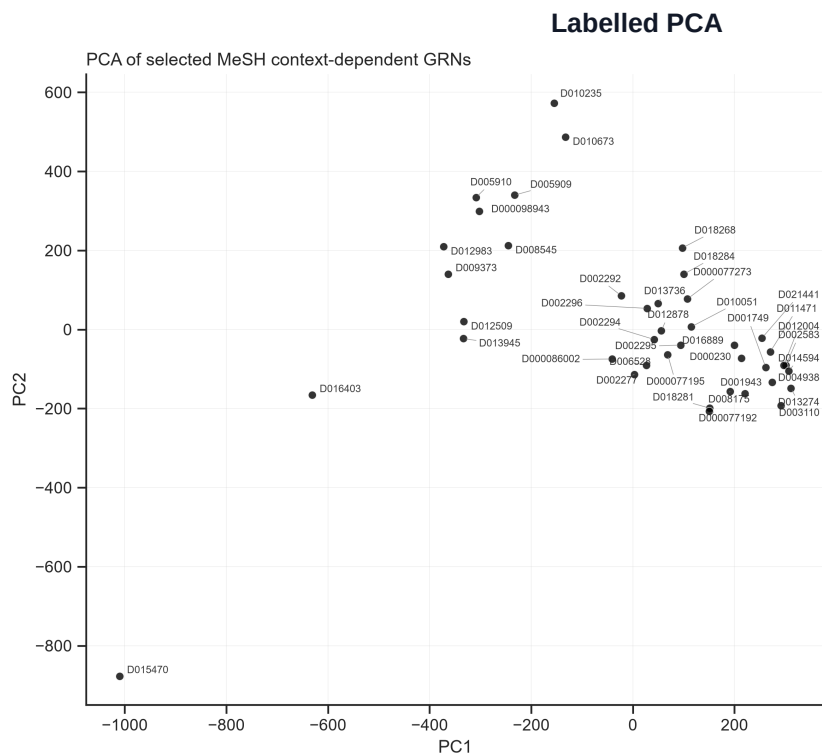
The results presented here provide an initial overview of the literature-derived disease networks and are intended to support subsequent biological assessment of their relationships and network-derived gene rankings.

TCGA-to-MeSH Mapping

TCGA	Cancer type	Mapped MeSH descriptor(s)	MeSH ID(s)
ACC	Adrenocortical Carcinoma	Adrenocortical Carcinoma	D018268
BLCA	Bladder Urothelial Carcinoma	Urinary Bladder Neoplasms; Carcinoma, Transitional Cell	D001749; D002295
BRCA	Breast Invasive Carcinoma	Breast Neoplasms; Carcinoma	D001943; D002277
CESC	Cervical Squamous Cell Carcinoma and Endocervical Adenocarcinoma	Uterine Cervical Neoplasms; Carcinoma, Squamous Cell; Adenocarcinoma	D002583; D002294; D000230
CHOL	Cholangiocarcinoma	Cholangiocarcinoma	D018281
COAD	Colon Adenocarcinoma	Colonic Neoplasms; Adenocarcinoma	D003110; D000230
DLBC	Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	Lymphoma, Large B-Cell, Diffuse	D016403
ESCA	Esophageal Carcinoma	Esophageal Neoplasms; Carcinoma	D004938; D002277
GBM	Glioblastoma Multiforme	Glioblastoma	D005909
HNSC	Head and Neck Squamous Cell Carcinoma	Squamous Cell Carcinoma of Head and Neck	D000077195
KICH	Kidney Chromophobe	Carcinoma, Renal Cell	D002292
KIRC	Kidney Renal Clear Cell Carcinoma	Carcinoma, Renal Cell	D002292
KIRP	Kidney Renal Papillary Cell Carcinoma	Carcinoma, Renal Cell	D002292
LAML	Acute Myeloid Leukemia	Leukemia, Myeloid, Acute	D015470
LGG	Brain Lower Grade Glioma	Glioma	D005910
LIHC	Liver Hepatocellular Carcinoma	Carcinoma, Hepatocellular	D006528
LUAD	Lung Adenocarcinoma	Adenocarcinoma of Lung	D000077192
LUSC	Lung Squamous Cell Carcinoma	Lung Neoplasms; Carcinoma, Squamous Cell	D008175; D002294
MESO	Mesothelioma	Mesothelioma, Malignant	D000086002
OV	Ovarian Serous Cystadenocarcinoma	Ovarian Neoplasms; Cystadenocarcinoma, Serous	D010051; D018284
PAAD	Pancreatic Adenocarcinoma	Carcinoma, Pancreatic Ductal	D021441
PCPG	Pheochromocytoma and Paraganglioma	Pheochromocytoma; Paraganglioma	D010673; D010235
PRAD	Prostate Adenocarcinoma	Prostatic Neoplasms; Adenocarcinoma	D011471; D000230
READ	Rectum Adenocarcinoma	Rectal Neoplasms; Adenocarcinoma	D012004; D000230
SARC	Sarcoma	Soft Tissue Neoplasms; Sarcoma	D012983; D012509
SKCM	Skin Cutaneous Melanoma	Skin Neoplasms; Melanoma	D012878; D008545
STAD	Stomach Adenocarcinoma	Stomach Neoplasms; Adenocarcinoma	D013274; D000230
TGCT	Testicular Germ Cell Tumors	Testicular Neoplasms; Neoplasms, Germ Cell and Embryonal	D013736; D009373
THCA	Thyroid Carcinoma	Thyroid Cancer, Papillary	D000077273
THYM	Thymoma	Thymoma	D013945
UCEC	Uterine Corpus Endometrial Carcinoma	Endometrial Neoplasms; Carcinoma	D016889; D002277
UCS	Uterine Carcinosarcoma	Uterine Neoplasms; Carcinosarcoma	D014594; D002296
UVM	Uveal Melanoma	Uveal Melanoma	D000098943

PCA and UMAP of Mapped MeSH GRNs

PCA and UMAP provide complementary views of the similarity among literature-derived disease networks. Nearby diseases share more similar network structures, whereas separated diseases exhibit greater differences in their literature-derived gene relationships. These visualisations are intended as an exploratory overview to guide subsequent biological interpretation.



Top 15 Genes Ranked by Literature-Derived GRN Importance

TCGA	MeSH descriptor	MeSH ID	Top 15 Genes Ranked by Literature-Derived GRN Importance
ACC	Adrenocortical Carcinoma	D018268	AKT1, CTNNB1, TP53, MYC, EGFR, ERBB2, TGFBI, MAPK1, NFKB1, ESR1, AR, MTOR, STAT3, CDH1, HIF1A
BLCA	Urinary Bladder Neoplasms	D001749	TP53, AKT1, CTNNB1, ERBB2, MYC, TGFBI, EGFR, MAPK1, NFKB1, CD8A, MTOR, ESR1, STAT3, CDH1, KRAS
BLCA	Carcinoma, Transitional Cell	D002295	TP53, AKT1, CTNNB1, TGFBI, EGFR, MYC, MAPK1, NFKB1, ERBB2, CDH1, STAT3, CD8A, AR, CD4, TNF
BRCA	Breast Neoplasms	D001943	TP53, AKT1, ERBB2, CTNNB1, MYC, EGFR, TGFBI, ESR1, NFKB1, MAPK1, MTOR, STAT3, CD8A, CDH1, VEGFA
BRCA	Carcinoma	D002277	TP53, AKT1, CTNNB1, MYC, TGFBI, NFKB1, MAPK1, EGFR, TNF, STAT3, ERBB2, CD8A, CDH1, CD4, HIF1A
CESC	Adenocarcinoma	D000230	TP53, AKT1, CTNNB1, EGFR, TGFBI, MAPK1, MYC, ERBB2, NFKB1, CDH1, ESR1, STAT3, AR, KRAS, MTOR
CESC	Carcinoma, Squamous Cell	D002294	TP53, AKT1, CTNNB1, TGFBI, MAPK1, EGFR, MYC, NFKB1, CDH1, ERBB2, STAT3, TNF, CD8A, CD4, ESR1
CESC	Uterine Cervical Neoplasms	D002583	TP53, AKT1, CTNNB1, ESR1, EGFR, ERBB2, MYC, TGFBI, NFKB1, MAPK1, CDH1, MTOR, VEGFA, CD8A, HIF1A
CHOL	Cholangiocarcinoma	D018281	TP53, AKT1, CTNNB1, TGFBI, MYC, NFKB1, MAPK1, EGFR, CDH1, STAT3, TNF, IL6, ERBB2, HIF1A, CD8A
COAD	Adenocarcinoma	D000230	TP53, AKT1, CTNNB1, EGFR, TGFBI, MAPK1, MYC, ERBB2, NFKB1, CDH1, ESR1, STAT3, AR, KRAS, MTOR
COAD	Colonic Neoplasms	D003110	TP53, AKT1, CTNNB1, MYC, NFKB1, TGFBI, MAPK1, EGFR, CDH1, TNF, STAT3, CD8A, MTOR, KRAS, ERBB2
DLBC	Lymphoma, Large B-Cell, Diffuse	D016403	TP53, MYC, CD4, AKT1, CD8A, NFKB1, CTNNB1, TNF, IFNG, TGFBI, BCL2, IL6, MAPK1, STAT3, IL10
ESCA	Carcinoma	D002277	TP53, AKT1, CTNNB1, MYC, TGFBI, NFKB1, MAPK1, EGFR, TNF, STAT3, ERBB2, CD8A, CDH1, CD4, HIF1A
ESCA	Esophageal Neoplasms	D004938	TP53, AKT1, CTNNB1, EGFR, MYC, TGFBI, NFKB1, MAPK1, CDH1, STAT3, ERBB2, TNF, ESR1, MTOR, CD8A
GBM	Glioblastoma	D005909	TP53, AKT1, APP, CTNNB1, TGFBI, MYC, EGFR, MAPK1, NFKB1, TNF, VEGFA, STAT3, HIF1A, MTOR, IL6
HNSC	Squamous Cell Carcinoma of Head and Neck	D000077195	TP53, AKT1, CTNNB1, TGFBI, EGFR, NFKB1, TNF, MAPK1, MYC, CDH1, ERBB2, STAT3, MTOR, CD8A, CD4
KICH	Carcinoma, Renal Cell	D002292	TP53, AKT1, CTNNB1, TGFBI, MAPK1, EGFR, MYC, NFKB1, AR, TNF, CDH1, CD8A, MTOR, VEGFA, STAT3
KIRC	Carcinoma, Renal Cell	D002292	TP53, AKT1, CTNNB1, TGFBI, MAPK1, EGFR, MYC, NFKB1, AR, TNF, CDH1, CD8A, MTOR, VEGFA, STAT3
KIRP	Carcinoma, Renal Cell	D002292	TP53, AKT1, CTNNB1, TGFBI, MAPK1, EGFR, MYC, NFKB1, AR, TNF, CDH1, CD8A, MTOR, VEGFA, STAT3
LAML	Leukemia, Myeloid, Acute	D015470	AKT1, TNF, TP53, NFKB1, MYC, CD4, TGFBI, CTNNB1, IL6, IFNG, STAT3, MAPK1, CD8A, IL1B, CD34
LGG	Glioma	D005910	TP53, AKT1, TGFBI, CTNNB1, MYC, NFKB1, EGFR, MAPK1, APP, TNF, STAT3, VEGFA, MTOR, HIF1A, IL6
LIHC	Carcinoma, Hepatocellular	D006528	TP53, AKT1, TGFBI, CTNNB1, MAPK1, MYC, NFKB1, TNF, EGFR, STAT3, HIF1A, CDH1, CD8A, MTOR, IL6
LUAD	Adenocarcinoma of Lung	D000077192	TP53, AKT1, TGFBI, EGFR, CTNNB1, MAPK1, MYC, NFKB1, ERBB2, ESR1, STAT3, CDH1, KRAS, TNF, MTOR
LUSC	Carcinoma, Squamous Cell	D002294	TP53, AKT1, CTNNB1, TGFBI, MAPK1, EGFR, MYC, NFKB1, CDH1, ERBB2, STAT3, TNF, CD8A, CD4, ESR1
LUSC	Lung Neoplasms	D008175	TP53, AKT1, EGFR, CTNNB1, MYC, TGFBI, NFKB1, MAPK1, MTOR, STAT3, CD8A, TNF, ERBB2, KRAS, CDH1
MESO	Mesothelioma, Malignant	D000086002	TP53, AKT1, TGFBI, MYC, CTNNB1, EGFR, NFKB1, MAPK1, CD8A, CDH1, MTOR, TNF, ERBB2, STAT3, CD274
OV	Ovarian Neoplasms	D010051	TP53, AKT1, CTNNB1, TGFBI, MYC, ERBB2, ESR1, EGFR, MAPK1, MTOR, NFKB1, CDH1, CD8A, VEGFA, STAT3
OV	Cystadenocarcinoma, Serous	D018284	TP53, AKT1, CTNNB1, EGFR, TGFBI, CDH1, ERBB2, MAPK1, MYC, NFKB1, AR, ESR1, VEGFA, STAT3, MTOR
PAAD	Carcinoma, Pancreatic Ductal	D021441	TP53, AKT1, CTNNB1, EGFR, TGFBI, ERBB2, NFKB1, MYC, MAPK1, AR, CD8A, ESR1, CDH1, STAT3, KRAS
PCPG	Paraganglioma	D010235	AKT1, TP53, CTNNB1, EGFR, MAPK1, TGFBI, MYC, NFKB1, MTOR, CDH1, AR, ERBB2, VEGFA, STAT3, TNF
PCPG	Pheochromocytoma	D010673	AKT1, CTNNB1, TP53, TGFBI, EGFR, NFKB1, MAPK1, MYC, MTOR, HIF1A, VEGFA, ERBB2, AR, PTEN, CD8A
PRAD	Adenocarcinoma	D000230	TP53, AKT1, CTNNB1, EGFR, TGFBI, MAPK1, MYC, ERBB2, NFKB1, CDH1, ESR1, STAT3, AR, KRAS, MTOR
PRAD	Prostatic Neoplasms	D011471	TP53, AKT1, CTNNB1, MYC, AR, TGFBI, EGFR, NFKB1, MAPK1, ERBB2, MTOR, STAT3, CDH1, PTEN, KRAS
READ	Adenocarcinoma	D000230	TP53, AKT1, CTNNB1, EGFR, TGFBI, MAPK1, MYC, ERBB2, NFKB1, CDH1, ESR1, STAT3, AR, KRAS, MTOR
READ	Rectal Neoplasms	D012004	AKT1, TP53, CTNNB1, MYC, EGFR, NFKB1, MAPK1, TGFBI, KRAS, AR, CDH1, VEGFA, STAT3, ERBB2, MTOR
SARC	Sarcoma	D012509	AKT1, TP53, TGFBI, CTNNB1, MYC, MAPK1, NFKB1, TNF, STAT3, VEGFA, MTOR, EGFR, IL6, HIF1A, MAPK3
SARC	Soft Tissue Neoplasms	D012983	AKT1, TP53, CTNNB1, TGFBI, NFKB1, MYC, MAPK1, TNF, CD4, EGFR, STAT3, CD8A, ALK, VEGFA, IL6
SKCM	Melanoma	D008545	TP53, AKT1, CTNNB1, TGFBI, MYC, MAPK1, NFKB1, EGFR, CD4, CD8A, VEGFA, TNF, MTOR, IFNG, STAT3
SKCM	Skin Neoplasms	D012878	TP53, AKT1, CTNNB1, MYC, EGFR, NFKB1, TGFBI, MAPK1, CD8A, MTOR, CD4, TNF, STAT3, ERBB2, VEGFA
STAD	Adenocarcinoma	D000230	TP53, AKT1, CTNNB1, EGFR, TGFBI, MAPK1, MYC, ERBB2, NFKB1, CDH1, ESR1, STAT3, AR, KRAS, MTOR
STAD	Stomach Neoplasms	D013274	TP53, AKT1, CTNNB1, MYC, EGFR, NFKB1, ERBB2, MAPK1, TGFBI, STAT3, KRAS, CDH1, MTOR, TNF, CD8A
TGCT	Neoplasms, Germ Cell and Embryonal	D009373	AKT1, TP53, CTNNB1, MYC, TGFBI, MAPK1, EGFR, NFKB1, TNF, MTOR, VEGFA, CD4, STAT3, HIF1A, CD8A
TGCT	Testicular Neoplasms	D013736	TP53, AKT1, CTNNB1, MYC, NFKB1, AR, EGFR, MAPK1, CD8A, TGFBI, MTOR, CD4, STAT3, CDH1, VEGFA
THCA	Thyroid Cancer, Papillary	D000077273	TP53, AKT1, CTNNB1, EGFR, MYC, TGFBI, MAPK1, NFKB1, ERBB2, CDH1, ESR1, AR, MTOR, STAT3, CD8A
THYM	Thymoma	D013945	TP53, TGFBI, AKT1, CD4, CD8A, CTNNB1, MYC, NFKB1, TNF, IFNG, EGFR, MAPK1, STAT3, IL6, ALK
UCEC	Carcinoma	D002277	TP53, AKT1, CTNNB1, MYC, TGFBI, NFKB1, MAPK1, EGFR, TNF, STAT3, ERBB2, CD8A, CDH1, CD4, HIF1A
UCEC	Endometrial Neoplasms	D016889	AKT1, TP53, ESR1, CTNNB1, EGFR, TGFBI, MYC, ERBB2, NFKB1, MAPK1, CDH1, VEGFA, MTOR, STAT3, TNF
UCS	Carcinosarcoma	D002296	AKT1, TP53, CTNNB1, TGFBI, MYC, NFKB1, EGFR, MAPK1, ERBB2, CDH1, HIF1A, MTOR, STAT3, CD8A, VEGFA
UCS	Uterine Neoplasms	D014594	TP53, AKT1, CTNNB1, ESR1, EGFR, MYC, ERBB2, TGFBI, NFKB1, MAPK1, CDH1, STAT3, VEGFA, HIF1A, MTOR
UVM	Uveal Melanoma	D000098943	AKT1, TP53, TGFBI, CTNNB1, MYC, MAPK1, VEGFA, BRAF, NFKB1, CD8A, TNF, CD4, EGFR, IFNG, STAT3

The genes listed above represent the highest-ranked genes within each literature-derived disease network. These rankings provide a concise reference for biological assessment of the current TCGA-MeSH mappings and comparison between disease-specific network profiles.